Unit: Unit 8: Quadratics **Name:** ____________
Topic: Systems of Linear and Quadratic Equations **Date:** ________

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A spacecraft is traveling through space with a velocity of $v = 2t + 5$. The position of the spacecraft is given by $s =$t^2 + 3t + $2. Find the time when the spacecraft is 50 km away from the starting point. Sho$

Q10. The distance between two planets is given by $d =$t^2 - 4t + $5, where d is the distance in astronomical units (AU) and t is time in years. A spacecraft is$
= 2t + $1. Find the maximum distance between the two planets and explain your answer. Show all work.$

Chef's Tips

- Check calculations carefully
- Ensure answers are labeled correctly
- Use a ruler to draw graphs